1 $\sin x = \frac{1}{\sqrt{2}}$ **[M1]**

$x = 45°$ **[B1]**

$x = 135°$ **[B1]**

[Total Marks: 3]

2 **(a)** $(180, -1)$ **[B1]**

(b) $60°$ **[B1]**

$300°$ **[B1]**

[Total Marks: 3]

3 **(a)** Smooth sine wave shape passing through $(0, 0)$, $(180, 0)$ and $(360, 0)$ **[B1]**

Maximum turning point at $(90, 1)$ and minimum turning point at $(270, -1)$ **[B1]**

(b) $x = 30$ **[B1]**

$x = 150$ **[B1]**

[Total Marks: 4]

4 **(a)** $\cos x = 0.5$ **[M1]**

$x = 60°$ **[B1]**

$x = 300°$ **[B1]**

(b) 0 (since $\sin x = \frac{5}{3} > 1$) **[B1]**

[Total Marks: 4]

5 **(a)** Smooth cosine wave starting at $(0°, 1)$, passing through $(90°, 0)$, reaching minimum at $(180°, -1)$ **[B1]**

Passing through $(270°, 0)$ and ending at $(360°, 1)$ with correct turning point symmetry **[B1]**

(b) Rearranges to $\cos x = -\frac{1}{2}$ **[M1]**

$120°$ **[A1]**

$240°$ **[A1]**

[Total Marks: 5]

6 **(a)** Smooth sine curve passing through $(0°, 0)$, $(180°, 0)$, and $(360°, 0)$ **[B1]**

Maximum point at $(90°, 1)$ and minimum point at $(270°, -1)$ **[B1]**

(b) Maximum $= (90°, 1)$ **[B1]**

Minimum $= (270°, -1)$ **[B1]**

(c) **(i)** $60°$ **[B1]**

$120°$ **[B1]**

(ii) $\cos x = -\frac{1}{2}$ **[M1]**

$120°$ **[A1]**

$240°$ **[A1]**

[Total Marks: 9]

7 **(a)** Smooth U-shaped cosine curve starting at $(0, 1)$ and ending at $(4, 1)$ representing $360°$ **[B1]**

Curve passes through $(1, 0)$, $(3, 0)$ and has a minimum at $(2, -1)$ **[B1]**

(b) $(180°, -1)$ **[B1]**

(c) $\cos x = -0.5$ (or $-\frac{1}{2}$) **[M1]**

$x = 120°$ **[B1]**

$x = 240°$ **[B1]**

[Total Marks: 6]

8 **(a)** Smooth sine curve passing through $(0, 0), (90°, 1), (180°, 0), (270°, -1), (360°, 0)$ **[B1]**

Smooth cosine curve passing through $(0, 1), (90°, 0), (180°, -1), (270°, 0), (360°, 1)$ **[B1]**

Intersection points shown approximately accurately at $x = 45°$ and $x = 225°$ **[B1]**

(b) $\tan x = 1$ **[M1]**

$x = 45°, 225°$ (Both required) **[A1]**

(c) Identifies the region between the two intersection points **[M1]**

$45° < x < 225°$ **[A1]**

[Total Marks: 7]

9 **(a)** $\sin x = -\frac{\sqrt{3}}{2}$ **[M1]**

$240°$ **[A1]**

$300°$ **[A1]**

(b) $\cos^2 x = \frac{3}{4}$ **[M1]**

$\cos x = \pm\frac{\sqrt{3}}{2}$ **[M1]**

$30°$ and $150°$ **[A1]**

$210°$ and $330°$ **[A1]**

(Award maximum of 3 marks if $\pm$ is omitted, resulting in only two solutions)

[Total Marks: 7]

10 **(a)** Dashed vertical asymptotes drawn at $x = 90°$ and $x = 270°$ **[B1]**

Correct increasing branches in $0° \leq x < 90°$ and $270° < x \leq 360°$ passing through $(0°, 0)$ and $(360°, 0)$ **[B1]**

Correct increasing branch in $90° < x < 270°$ passing through $(180°, 0)$ **[B1]**

(b) $90°$ **[B1]**

$270°$ **[B1]**

(c) **(i)** $\tan x = -\dfrac{1}{\sqrt{3}}$ **[M1]**

$150°$ **[A1]**

$330°$ **[A1]**

(ii) $(2\sin x + 1)(\sin x - 1) = 0$ **[M1]**

$\sin x = -\dfrac{1}{2}$ or $\sin x = 1$ **[A1]**

$x = 90°$ **[B1]**

$x = 210°$ and $x = 330°$ **[A1]**

[Total Marks: 12]

11 **(a)** Sine wave passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ **[B1]**

Smooth curve with a maximum at $(90, 1)$ and a minimum at $(270, -1)$ **[B1]**

(b) $\arcsin(0.7)$ OR $44.42\ldots$ **[M1]**

$44.4°$ **[A1]**

$135.6°$ **[A1]**

[Total Marks: 5]

12 **(a)** $\sin x = -0.75$ **[M1]**

Reference angle $= 48.6°$ (or $48.59...°$) **[M1]**

$x = 228.6°, 311.4°$ (Both required for the final mark) **[A1]**

(b) x-coordinate is $270°$ **[B1]**

y-coordinate is -1 **[B1]**

[Total Marks: 5]

13 **(a)** Correct basic shape with multiple branches passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ **[B1]**

Asymptotes at $x = 90$ and $x = 270$ clearly indicated or graph showing asymptotic behaviour **[B1]**

(b) $\tan x = -\frac{5}{3}$ or $-1.66\ldots$ **[M1]**

$121.0°$ **[A1]**

$301.0°$ **[A1]**

[Total Marks: 5]

14 (a) $\cos x = -0.25$ **[M1]**

$x = 104.5$ **[A1]**

$x = 360 - 104.47... = 255.5$ **[A1]**

(b) $\tan x = 2.5$ **[M1]**

$x = 68.2$ **[A1]**

$x = 68.19... + 180 = 248.2$ **[A1]**

[Total Marks: 6]

15 (a) Finds acute reference angle: $\sin^{-1}(0.64) = 39.79...°$ **[M1]**

$219.8°$ **[A1]**

$320.2°$ **[A1]**

(b) Rearranges to $\tan x = -\frac{5}{3}$ and finds reference angle $\tan^{-1}\left(\frac{5}{3}\right) = 59.036...°$ **[M1]**

$121.0°$ (accept $121°$) **[A1]**

$301.0°$ (accept $301°$) **[A1]**

[Total Marks: 6]

16 (a) $\cos x = -\frac{1}{3}$ **[M1]**

$x = 109.5°$ (accept $109.47\ldots$) **[A1]**

$360° - 109.47°$ **[M1]**

$x = 250.5°$ (accept $250.52\ldots$) **[A1]**

(b) Identifies that intersection with x-axis means $y = 0$, so x-coordinates are the solutions from part (a) **[M1]**

Distance $= 250.53° - 109.47°$ **[M1]**

141.1 (accept 141.0 to 141.1) **[A1]**

[Total Marks: 7]

17 (a) $\cos^{-1}(-0.35) = 110.487\ldots°$ **[M1]**

x-coordinate of $A = 110.5°$ **[A1]**

x-coordinate of $B = 360° - 110.487\ldots° = 249.5°$ **[A1]**

(b) $\sin x = -0.6$ **[M1]**

$216.9°$ **[A1]**

$323.1°$ **[A1]**

(c) $\tan x = \frac{7}{3}$ **[M1]**

$66.8°$ **[A1]**

$246.8°$ **[A1]**

[Total Marks: 9]

18 (a) $\sin 30° = \frac{1}{2}$ leading to $a\left(\frac{1}{2}\right) - b = 0$ or $\frac{b}{a} = \frac{1}{2}$ **[M1]**

$a = 2$ and $b = 1$ **[A1]**

(b) 150 **[B1]**

(c) $k = \cos(242°) = -0.469$ **[B1]**

Uses symmetry of cosine curve: $360° - 242°$ **[M1]**

$x = 118°$ **[A1]**

[Total Marks: 6]

19 (a) $\tan x = -2.8$ **[M1]**

$x = 109.7°$ (accept $109.6°$ to $109.7°$) **[B1]**

$x = 289.7°$ (accept $289.6°$ to $289.7°$) **[B1]**

(b) Factorises to $\sin x(2\sin x - 1) = 0$ **[M1]**

Identifies $\sin x = 0$ or $\sin x = 0.5$ (or $\frac{1}{2}$) **[M1]**

$x = 0°, 180°, 360°$ (all three correct for the mark) **[A1]**

$x = 30°, 150°$ (both correct for the mark) **[A1]**

[Total Marks: 7]

20 $5\sin x = 2\frac{\sin x}{\cos x}$ OR $5\sin x\cos x = 2\sin x$ **[M1]**

$\sin x(5\cos x - 2) = 0$ [M2] (Award [M1] for correctly gathering terms on one side: $5\sin x\cos x - 2\sin x = 0$)

$\sin x = 0$ **[B1]**

$0°, 180°, 360°$ [A1] (Must have all three solutions for the mark)

$\cos x = 0.4$ **[B1]**

$66.4°$ (accept $66.42\ldots$) **[A1]**

$293.6°$ (accept $293.57\ldots$) **[A1]**

[Total Marks: 7]

21 **(a)** Divides by $\cos x$ to obtain $\tan x = \frac{2}{5}$ (or 0.4) **[M1]**

$21.8°$ **[A1]**

$201.8°$ **[A1]**

(b) Factorises quadratic in $\sin x$: $(2\sin x - 1)(\sin x + 1) = 0$ **[M1]**

Solves linear factors: $\sin x = \frac{1}{2}$ and $\sin x = -1$ **[M1]**

$30°$ **[A1]**

$150°$ **[A1]**

$270°$ **[A1]**

[Total Marks: 8]